TITLE: ML Classification Project using Titanic Dataset

1. Objective

   The objective of this project is to build and evaluate machine learning classification models to predict passenger survival on the Titanic dataset.

2. Dataset

   Dataset Used: Titanic Dataset

   Source: Titanic passenger records containing information such as age, sex, fare, passenger class, and survival status.

3. Data Preprocessing

   • Loaded dataset using Pandas

   • Checked missing values

   • Filled missing values in Age and Embarked columns

   • Removed Cabin column due to excessive missing values

   • Converted categorical variables (Sex, Embarked) into numerical values

   • Split dataset into training and testing sets

4. Models Used

    • Logistic Regression

    • Random Forest Classifier

5. Results

    Logistic Regression Accuracy: 81.00%

    Random Forest Accuracy: 82.12%

    ROC-AUC Score: 0.8893

6. Evaluation Metrics

    • Accuracy

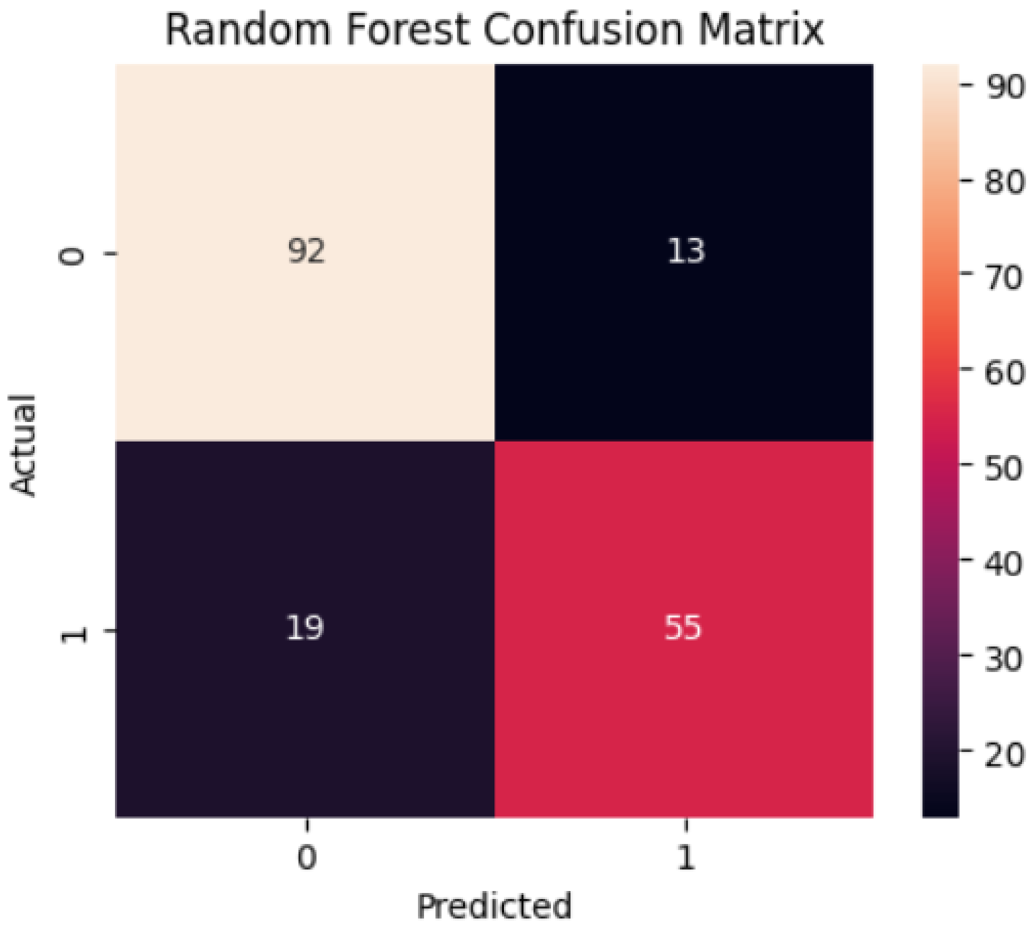
    • Precision

    • Recall

    • F1-Score

    • ROC-AUC

7. Confusion Matrix



8. Conclusion

The Random Forest model performed better than Logistic Regression with an accuracy of 82.12% and ROC-AUC score of 0.8893. Therefore, Random Forest was selected as the final model.